

Yelene Cisse

(267) 269-0578 | ymc2123@columbia.edu | linkedin.com/in/yelene-cisse | <https://ymciss0.github.io/>

EDUCATION

Columbia University

New York, NY

Master of Science in Data Science

Expected Dec 2026

Relevant courses: Applied Machine Learning, Algorithms for Data Science, Probability & Statistics, Data Analysis & Visualization.

La Salle University

Philadelphia, PA

Bachelor of Science in Mathematics

Aug 2018 - May 2022

Minors: Computer Science, Chemistry.

Honors and Awards.

2022 Outstanding Graduate of the Mathematics department, Maxima Cum Laude, Founder's Scholarship.

WORK EXPERIENCE

Chubb Ltd

Philadelphia, PA

Data Scientist II

Jul 2022 - Jul 2025

- Facilitated company-wide programming language transition, converting legacy SAS codebase to Python to ensure business continuity for actuarial pricing models.
- Optimized data pipeline managing 10 external sources with automated extraction, preprocessing, and feature engineering, improving model testing accuracy and computational efficiency.
- Built Generalized Linear models to inform actuarial pricing decisions in Small Commercial line of business, including Property, Auto and Worker's Compensation.
- Designed and presented analytical findings to both technical and non-technical stakeholders using PowerPoint, Excel, and QlikView Dashboards.
- Expedited production deployment by creating implementation tables and governance documentation for model auditing and future refresh cycles.
- Mentored an intern during development of a Natural Language Processing project aimed to build a classification model for Cyber claims categorization automation.
- Applied agile methodologies by leveraging JIRA for project tracking and GitHub for collaboration with teammates.

RELEVANT EXPERIENCE

SkinWise: AI-Powered Skincare Recommendation System

New York, NY

AML Team Project - Sentiment Analysis Lead

Dec 2025

- Developed a sentiment classification system using TF-IDF and logistic regression to analyze 581K Sephora reviews, achieving 86% accuracy for personalized product recommendations.
- Improved critical feedback detection by 46% (negative precision: 61% → 88%), ensuring the AI accurately identifies product issues and customer dissatisfaction.
- Designed review quality scoring algorithm, automatically selecting top-5 representative reviews per product and optimizing information delivery while maintaining recommendation quality.
- Validated multi-platform scalability by successfully applying trained model to 4K unlabeled Ulta reviews, demonstrating capability to ingest data from diverse sources (Reddit, blogs, e-commerce).

Bank Marketing Campaign Prediction

New York, NY

Applied Machine Learning course

Nov 2025

- Preprocessed 41K+ observations with 62 features from the UCI Bank Marketing dataset leveraging Python (pandas, scikit-learn, XGBoost).
- Developed and tuned ensemble models (Decision Tree, Random Forest, GBM, XGBoost) with GridSearchCV and 5-fold CV to address severe class imbalance (11.3% positive).
- Optimized Gradient Boosting achieving 36.9% F1-score and 46.4% AUCPR—22% improvement over Random Forest baseline.
- Presented results through comprehensive report, including precision-recall and validation curves, feature importance plots, and actionable insights for campaign optimization.

SKILLS

- Programming Language: Python (Pandas, NumPy, scikit-learn, H2O), SQL, R (ggplot2).
- Cloud Ecosystem: Databricks, ADLS.
- Data Visualization: Excel, QlikView, Matplotlib, Seaborn, Altair.
- Agile: Jira, GitHub.
- Language: French (Native), Spanish (Conversational).